November 12, 1999

Inorganic Cume

Answer **<u>either</u>** 1-3 **<u>or</u>** 4-7

1 – Identify and draw examples of all the isomers possible for a six-coordinate, tris unsymmetrical chelate complex .
Use the complex $Co(SCH_2CH_2NH_2)_3$ as your example.
Abbreviate the chelate ligand = $H_2\mathbf{N}CH_2CH_2\mathbf{S}^{1-}$ as

2 – Discuss and illustrate both the dissociative and non-dissocative mechanisms that can lead to interconversion of the $Co(SCH_2CH_2NH_2)_3$ isomers.

3 – Conversion of a single isomer into <u>all</u> of the other possible isomers may require more than one isomerization step. For **one** of the mechanisms discussed question 2, pick and draw a **single starting isomer** and discuss the complete set of isomeric products that can be obtained in **one step** by your chosen mechanism.

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4 – As prepared in the reaction of $Fe(CO)_5$ with dicyclopentadiene, $[Fe(Cp)(CO)_2]_2$ consists of two isomers. Each of the isomers has $\nu(CO)$ bands in both the 2000 and 1800 cm^{-1} regions.
<u>Identify the isomers and justify your structures on bonding and experimental grounds.</u> **Cp = $C_5H_5^{1-}$**

5- Variable temperature ^1H nmr for a mixture of the two isomers of $[Fe(Cp)(CO)_2]_2$ shows a single Cp resonance at higher temperature that broadens at – 48 °C and then splits into two ~equal intensity resonances at – 66 °C.
<u>Account for this observation and suggest a reasonable mechanism for the reaction.</u>

6 – The ^{13}C nmr spectra of the isomers in the CO ligand region also show interesting variable temperature changes. The more polar isomer shows two ^{13}C carbonyl resonances at <u>low</u> temperature but only one resonance at higher temperatures. The coalescence temperature behavior for this isomer is very similar to that observed for the Cp proton resonance experiments. (I.e. the E_a values for the processes are similar.)
<u>Is your mechanism proposed in #5 consistent with this observation?</u>

7 - The coalescence temperature for the less polar isomer is lower than that for the other isomer suggesting a lower activation barrier for CO equilibration.
<u>How is this observation consistent with your mechanism?</u>